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## **Revolutionizing Conventional Drilling: Cost- Effective Coil Tubing Drilling (CTD) for Deviated and Complex Well Designs - Feasibility & Readiness at a Glance**

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### **Abstract**

A marginal field off the coast of East Malaysia has a low recovery factor, yet it holds a substantial amount bypassed hydrocarbons within the shallow high-angle and depleted reservoirs, conditions that make conventional drilling economically unfeasible. Coiled Tubing Drilling (CTD) was identified as a cheaper alternative to access these reserves. However, the requirement for a dedicated vessel to reach the remote unmanned platforms significantly increases project costs and operational risks. To tackle this challenge, the project team implemented a phased strategy that separate the preparatory activities from the drilling campaign. This approach aimed to reduce NonProductive Time (NPT), enhance the efficiency of vessel utilization and de-risk operations.

The preparation phase began with a multidisciplinary subsurface assessment to select suitable candidates. Screening criteria included reservoir quality, remaining reserves, well integrity, well/platform accessibility, artificial lift consideration/issue and future intervention potential. Once candidates were selected, a detailed feasibility study was conducted to assess operational viability. This study encompassed platform site visits to evaluate space and lifting constraints, equipment layout planning and readiness assessments. In addition, technical reviews addressed fluid system design tailored to reservoir conditions, completion and cementing strategies, and trajectory optimization to ensure maximum reservoir contact and optimize well productivity while mitigating operational risks.

This structured and integrated approach highlights how CTD can be systematically applied in Malaysian offshore marginal fields to deliver technically robust, cost-efficient and operationally safe outcomes, while ensuring alignment with broader long-term field development goals. The paper further discusses lessons learned, best practices and forward plans from these campaigns to guide future CTD applications in similar environments.

### **Introduction**

Balingian province is a proven hydrocarbon province in Sarawak that has been actively explored since the early 1920s and to date has been producing from at least seven (7) moderately sized oil fields (Roy et al.,

2010). Field X located approximately 30 km west of Bintulu, one of the earliest fields in the province, was discovered in 1962 and brought into production in 1979. It was a major producer, from seven (7) unmanned platforms strategically located around the field, with production reported at over 9,000 barrels per day (bpd) around the year 2000. Geologically, it is situated on a structural high or ridge within the Balingian subbasin. The field's subsurface architecture is highly complex, characterized as a westerly plunging fold structure bounded by two (2) major reverse faults to the north and southeast, which converge to the northeast and diverge to the southwest. This extensive and varied faulting, while posing significant challenges for reservoir characterization and fluid flow, also creates numerous trapping mechanisms and the potential for bypassed oil. The inherent geological complexities of the field (i.e. extensive faulting and unconsolidated reservoirs), combine with the natural pressure decline, requiring continuous intervention and innovation to unlock remaining reserves while managing technical, operational and regulatory risks.

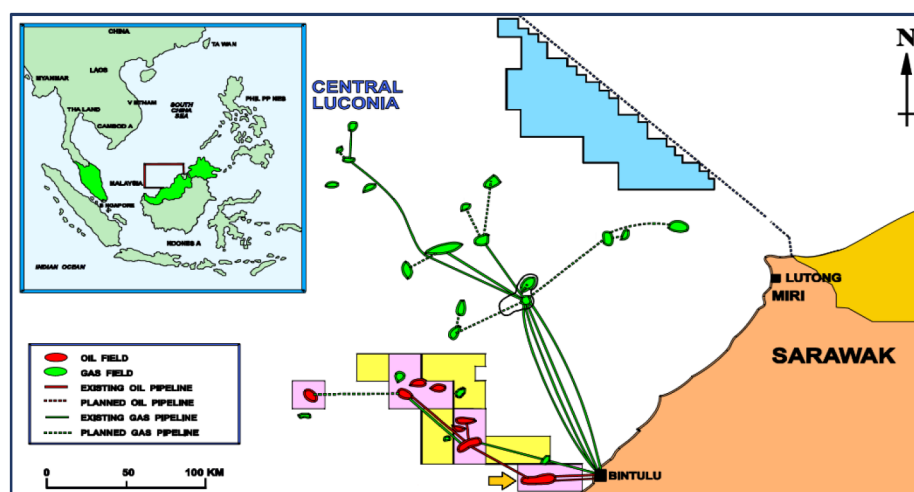


Figure 1—Field X Location Map (Ali K. et al., 2006)

Field X is classified as a "*brownfield*", a designation for mature fields that have been producing for an extended period and require continuous efforts to maintain or enhance production. Despite its long history, Field X remains strategically vital and holds significant hydrocarbon potential, necessitating continuous initiatives to maximize its value (i.e. the recovery factor remained at 21% and is targeting an increase to 54%). These initiatives include Production Enhancement and Idle Well Reactivation (PE-IWR), Improved Oil Recovery (IOR) (e.g. water injection, dump flooding) and the application of advanced subsurface technologies.

However, operating this mature field poses several technical and economic challenges that complicate ongoing development. Sand production remains a critical concern, as it can cause severe damage to both downhole and surface equipment, leading to increased maintenance and intervention costs. Aging infrastructure has also introduced emerging well integrity risks, while depleted and shallow reservoirs are increasingly difficult to access with conventional recovery methods. Beyond these subsurface challenges, reservoir complexity, particularly in terms of connectivity and fluid contact uncertainties, adds further unpredictability to field performance. Regulatory and environmental constraints (i.e. PETRONAS Net Zero Carbon Emission by 2050 Pathway), impose additional limitations on operational flexibility, particularly with respect to routine gas flaring. Furthermore, nearly half of the wells in the field are currently idle, representing both a challenge in terms of lost production and an opportunity for re-entry to unlock remaining reserves. These wells can serve as good candidates (i.e. "*donor well*") for PE-IWR initiative (e.g. CTD, workover, etc.) offering a cost-effective way to access remaining reserves without drilling entirely new wells.

In this context, CTD has been identified as the preferred approach to extract the remaining hydrocarbons in Field X. CTD offers several advantages, including a smaller footprint, the ability to perform through-tubing sidetracks, the potential to operate underbalanced to preserve reservoir productivity and the ability to navigate challenging well geometries. Building on the foundations laid by the 2011 pilot and the 2013 Self Elevated Workover Platform (SEWOP) campaign, the current CTD project represents the next stage in Malaysia's CTD journey. Leveraging the lessons from earlier phases, particularly in candidate well selection, logistics optimization and equipment reliability, the project team refined the workflow by decoupling preparatory activities from the drilling campaign. This approach aimed to further reduce non-productive time (NPT), optimize vessel utilization and sustainable field development.

## Lessons from Previous CTD Campaigns in Malaysia

The early CTD campaigns in Malaysia provided a valuable foundation for shaping subsequent projects, where key lessons learned are summarized below:

- **Through-Tubing Feasibility Proven:** The 2011 pilot confirmed that coiled tubing could successfully drill sidetracks from existing completions, providing a cost-effective alternative to conventional rig-based approaches.
- **Integrated Workflow Required:** The full project scope, from plug and abandonment (P&A) to whipstock setting, window milling, lateral drilling and completion, demonstrated the need for a structured multidisciplinary workflow.
- **Logistics Drive Success:** The 2013 campaign showed that logistics solutions such as the SEWOP can overcome space and load limitations on mature offshore platforms, reducing modification requirements and minimizing weather-related delays.
- **Continuous Improvement of Tools and Processes:** Early challenges with downhole tool reliability and platform preparation underscored the importance of robust maintenance practices, contingency planning and ongoing refinement of CTD equipment and methods.
- **Economic Viability Confirmed:** Both campaigns demonstrated that, under the right conditions, CTD is not only technically feasible but also economically competitive for mature and marginal field development.



Figure 2—CTD Lessons Learnt

A critical success factor for the CTD campaign in Field X was the robustness of the planning phase. A comprehensive design and feasibility assessment was undertaken to ensure that each candidate well was technically viable, economically justifiable and operationally safe. All potential candidates underwent a rigorous screening process, which include laboratory testing of reservoir fluids and materials to validate

completion design parameters and confirm compatibility with the existing well architecture and reservoir conditions. The outcome of this step ensured that the selected designs would not compromise long-term well integrity or production performance. Furthermore, a Project Risk Assessment (PRA) was conducted to systematically identify potential hazards throughout the CTD campaign. Mitigation measures were incorporated into the program design, covering risks such as well control, equipment reliability, weather delays and sand production. To further strengthen operational readiness, Wells on Paper (WOP) exercises were organized with multidisciplinary teams, simulating the execution sequence and testing the decision-making process under various contingency scenarios. These pre-operational assessments, not only validated design assumptions but also aligned all stakeholders (e.g. subsurface, drilling, operations, HSE teams) on procedures, responsibilities and contingency actions. This structured and collaborative approach provided a solid foundation for safe, efficient and technically sound CTD execution in Field X.

## Methodology

The methodology for the CTD campaign in Field X was developed to ensure technical feasibility, operational efficiency and long-term well performance. The workflow was structured into six (6) major components: (1) Candidate Selection, (2) Facilities Readiness, (3) Trajectory Design, (4) Fluid Design, (5) Completion Design, and (6) Method of Operations.

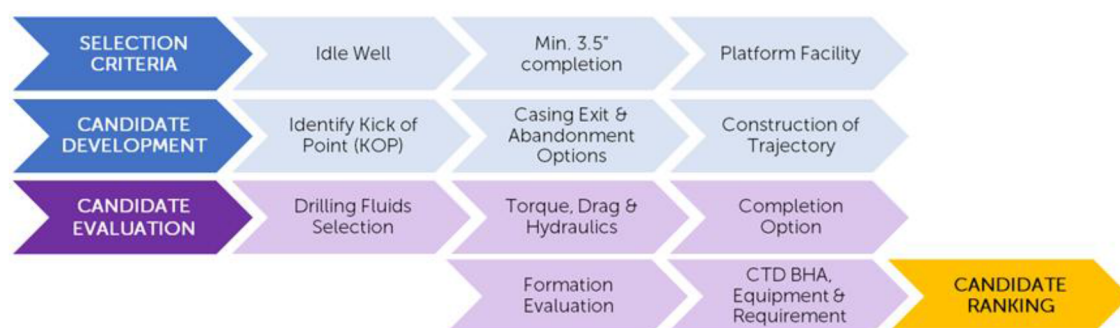


Figure 3—CTD Candidate Selection

## Candidate Selection

The candidate selection process for CTD follows principles similar to conventional PE-IWR projects (e.g. sidetrack, workover, etc.), but with added constraints specific to Coiled Tubing (CT) technology. The process began with identification of suitable reservoir targets, primarily based on the volume of remaining reserves. However, reserve size alone was insufficient, the quality of the reservoir rock plays a critical role in determining the feasibility and potential success of the operation. High-quality reservoir rock ensures better permeability and porosity, which are essential for efficient hydrocarbon recovery.

Before finalizing the well trajectory for CTD, a detailed iterative process involving the geological model is conducted. This was essential to optimize reservoir drainage while minimizing operational risks. The trajectory must be carefully planned to maximize contact with the productive zones without compromising well integrity or increasing the likelihood of drilling complications (e.g. unnecessary tortuosity, high dog-leg severity). In Field X, although nearly half of the well inventory is currently idle, selecting a suitable *donor well* is far from straightforward. The age of the wells, many of which were drilled over thirty years (30) ago, raises significant concerns regarding well integrity and accessibility. These factors must be thoroughly evaluated before any CTD operation can proceed. Additionally, some wells present unique challenges due to wax deposition at shallow depths, which can obstruct access and complicate intervention efforts.

As a result, the process demands a more nuanced and cautious approach. It requires a balance between geological potential and operational practicality, ensuring that the selected wells not only offer promising



reservoir targets but are also technically viable for well intervention (especially with CT), which includes several key criteria listed below:

- Minimum 3-½" production tubing, either single string or dual strings
- Subsurface target within approximate 3,000ft from the *donor well*
- Minimum two (2) wells candidates in single platform

### Facilities Readiness

CTD operations in offshore environments present a unique challenges, particularly in terms of equipment setup and the method of conveying the CT. Unlike onshore operations, offshore platforms often have spatial and structural limitations that require careful planning and adaptation. A key decision in planning is whether the CTD unit can be fully accommodated on the platform or must be operated via catenary support from a vessel/workboat.

In Field X, all wells are located on remote offshore platforms that have limited deck space and lifting capabilities. The catenary CTD method was therefore selected as the only viable option. This approach deployed CT from a vessel/workboat, with the tubing suspended in a curved configuration between vessel/workboat and platform. The use of a support vessel/workboat adds complexity to the logistics and coordination of the drilling campaign, necessitating detailed planning and risk assessment.



Figure 4—Previous CTD Operation on Platform X

Pre-job platform site visits were conducted to ensure the success of the CTD campaign. These site visits were essential for identifying platform-specific limitations, developing tailored equipment layouts and establishing lifting plans. Both pre-CTD and CTD operations were considered during this planning phase to ensure smooth execution and minimize operational risks.

Another critical aspect of the CTD campaign is the readiness of platform facilities post-drilling, where once the CTD operation is completed the well must be unloaded for production. Simultaneous Operations (SIMOPS) are not allowed in Field X, therefore well unloading has to be done offline. This delay can negatively impact productivity, as the prolonged presence of drilling/completion fluids in the wellbore may reduce reservoir performance (i.e. productivity loss due to prolonged foreign fluid residence in the wellbore). In order to lessen back pressure and detrimental effect on productivity, a bespoke unloading strategy was developed for each well that involves using Nitrogen (N<sub>2</sub>) to lift a portion of the fluid column onto surface tanks for temporary storage. This approach helps preserve reservoir integrity and production performance, even in the absence of SIMOPS.

Overall, CTD campaign in Field X require a highly customized approach that accounts for offshore platform limitations, logistical challenges and post-drilling constraints. Through careful planning,

innovative methods like catenary support and bespoke unloading strategy and thorough site assessments, the project team can effectively navigate these complexities and optimize well performance.

## Trajectory Design

Following the selection of *donor well* and the identification of subsurface targets, a detailed evaluation is conducted to determine the optimal path from the existing completion to the new reservoir targets. This step is critical in ensuring that the CTD sidetrack can be executed efficiently and safely. Several factors are considered during this phase, as listed below:

- Complexity of existing completion and restrictions in the motherbore
- Spatial relationship between the subsurface targets and the motherbore
- Capabilities of the Bottom Hole Assembly (BHA) and planned lower completion

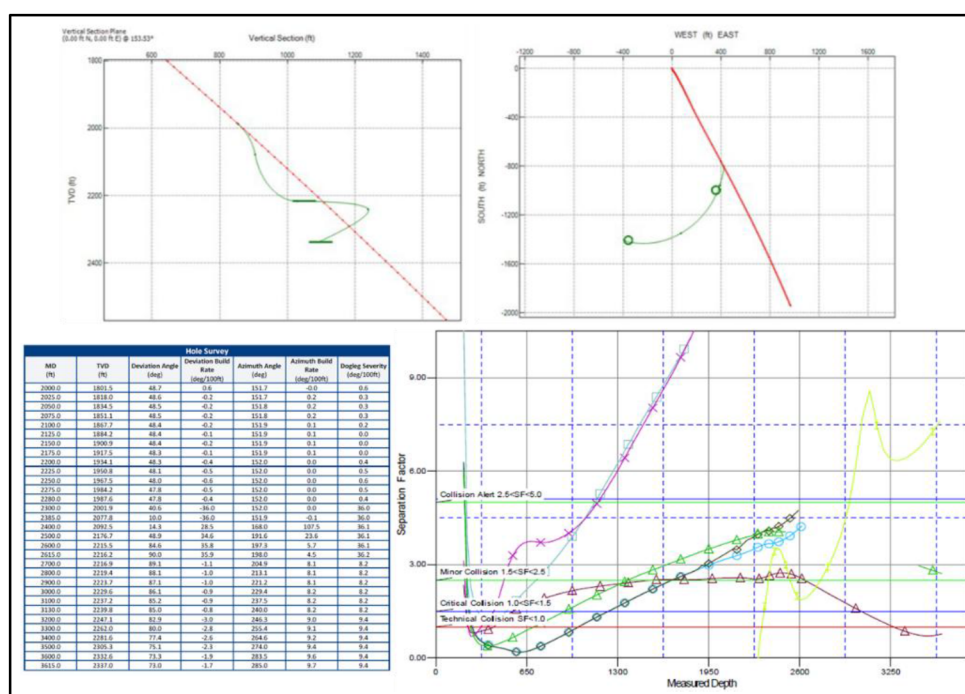


Figure 5—Field X Well Trajectory Planning

A key aspect of trajectory planning is determining the ideal placement of the whipstock. The goal is to initiate the sidetrack as close as possible to the target zone, minimizing the openhole length of the non-productive zones and reducing wellbore tortuosity. The direction of the exit is equally important, as it must be compatible with both the whipstock design and milling capabilities to ensure a smooth and controlled kick-off.

Once the preliminary trajectory is defined, further technical evaluations are performed, particularly torque-drag analysis and hydraulic simulations. These assessments are essential to confirm that the proposed trajectory is within the operational limits of the CTD and aligns with previous experiences (i.e. historical CTD performance). By integrating mechanical feasibility with reservoir targets, the designs optimized both drilling efficiency and long-term productivity.

## Fluid Design

Fluid system selection is a critical component in the overall feasibility study for CTD, particularly given slimhole constraints and offshore environments like Field X. The choice of drilling and completion fluids

will directly impact several key operational and reservoir performance factors, including fluid rheology, borehole cleaning efficiency, weight transfer from the CT to the bit and borehole stability. Given the unique constraints of CTD (e.g. the inability to rotate the tubing, the need for wiper trips, etc.), fluid design must be carefully tailored to ensure effective drilling while minimal formation damage.

Several technical considerations must be addressed during fluid system selection. First, increased friction pressures due to the small Internal Diameter (ID) of the CT may have an impact on weight transfer and fluid circulation. Additionally, the increased Equivalent Circulating Density (ECD) caused by reduced annular clearance between CT and the wellbore may influence borehole stability and impact on the fracture gradient. These factors must be balanced to avoid inducing formation fractures or compromising well integrity. Compatibility with the reservoir, completion equipment and surface facilities also essential to prevent screen plugging, corrosion risk or other issues that could impair productivity, operational efficiency (e.g. unplanned shut-down due to severe emulsion during well unloading) and Asset integrity (e.g. corrosion of surface flowline, damage of existing upper completion).

For Field X, a non-aqueous synthetic-based drilling and completion fluid was selected with a premix density between 7.4 to 8.0 pounds per gallon (ppg). This formulation helps mitigate the risk of swelling clays due to sensitive formation mineralogy and minimizes formation damage during the openhole drilling phase. The synthetic-base fluid is also logistically and economically favorable, as it can be reused with minimal treatment, hence reducing waste and operational costs. Fluid displaced in the openhole section were designed to be solid-free to support the completion operation while maintaining borehole stability. Subsequently, a higher-density fluid is displaced inside the casing to provide the necessary Equivalent Static Density (ESD) during the installation of the completion assemblies.

Table 1—Fluid System Properties

| Properties              | Specification |
|-------------------------|---------------|
| Density, ppg            | 9.3 – 9.7     |
| PV, cP                  | 10-20         |
| YP, lb/100sqft          | 10 - 20       |
| Gel 10s/10min           | >6/<20        |
| HTHP F/L, ml/30min      | <6            |
| CaCl <sub>2</sub> , %wt | 24% – 27% wt  |
| ESV                     | 200 – 400     |
| LGS, %                  | <2%           |

To validate the fluid design, a series of laboratory tests were conducted to assess reservoir compatibility, completion equipment interaction and potential impacts on well productivity. These evaluations ensure that the selected fluid system supports both drilling performance and long-term reservoir/well productivity. The key tests and their objectives are described below

#### ***Reservoir Drill-In Fluid (RDIF) Design.***

- The RDIF was formulated to deliver the dual objectives of efficient hole cleaning and minimal formation damage.
- Laboratory testing evaluated rheological properties, lubricity and carrying capacity under simulated downhole pressure and temperature.
- Focus was placed on optimizing particle size distribution (PSD) of bridging materials to achieve effective filtercake deposition while ensuring easy removal during clean-up.

#### ***Breaker Tests.***

- Breaker fluids were assessed for compatibility with crude oil, flowback efficiency and potential corrosion risk.
- The tests confirmed that the breaker could effectively degrade filtercake residues and restore permeability without inducing adverse emulsions or corrosion on tubular materials.
- Static soak and dynamic flow tests were performed to replicate expected downhole exposure times.

#### ***Fluid Compatibility with Reservoir Core Samples.***

- Core samples from Field X were used to evaluate interactions between drilling/completion fluids and formation rock.
- Tests addressed risks of formation damage such as scale precipitation, wax deposition, and emulsion formation.
- The assessments guided adjustments in fluid chemistry to ensure stable wettability and to avoid permeability impairment under reservoir conditions.

#### ***Production Screen Tests (PST).***

- To evaluate screen plugging potential, fluids were flowed through representative stand-alone screen samples under controlled differential pressure.
- Tests quantified solids retention and pressure drop across the screen to confirm long-term sand exclusion without excessive plugging risk.
- Results provided validation for the selected screen mesh size (200 gm) in conjunction with the fluid system.

#### ***Swell Packer Tests.***

- Swellable elastomer packers were exposed to the selected fluid systems under reservoir temperature and pressure conditions.
- Tests measured mechanical integrity, swelling rate, and sealing performance to ensure reliable zonal isolation once deployed downhole.
- Additional cycling tests simulated thermal and pressure fluctuations expected during the life of the well.

#### ***Retained Permeability Tests (RPT).***

- RPTs were conducted on reservoir core plugs to quantify the impact of fluid invasion and clean-up on effective permeability (i.e. to measure productivity impact).
- Results measured the percentage of permeability retained after exposure to RDIF and subsequent breaker treatment.
- High retained permeability values (>80%) confirmed that the selected fluid and breaker system would minimize formation damage and sustain long-term productivity.

### **Completion Design**

Designing the completion for CTD wells in Field X required careful attention to constraints imposed by the existing well architecture, reservoir conditions and operational history. The key considerations were as follows:

#### ***Wellbore Geometry and Restrictions.***

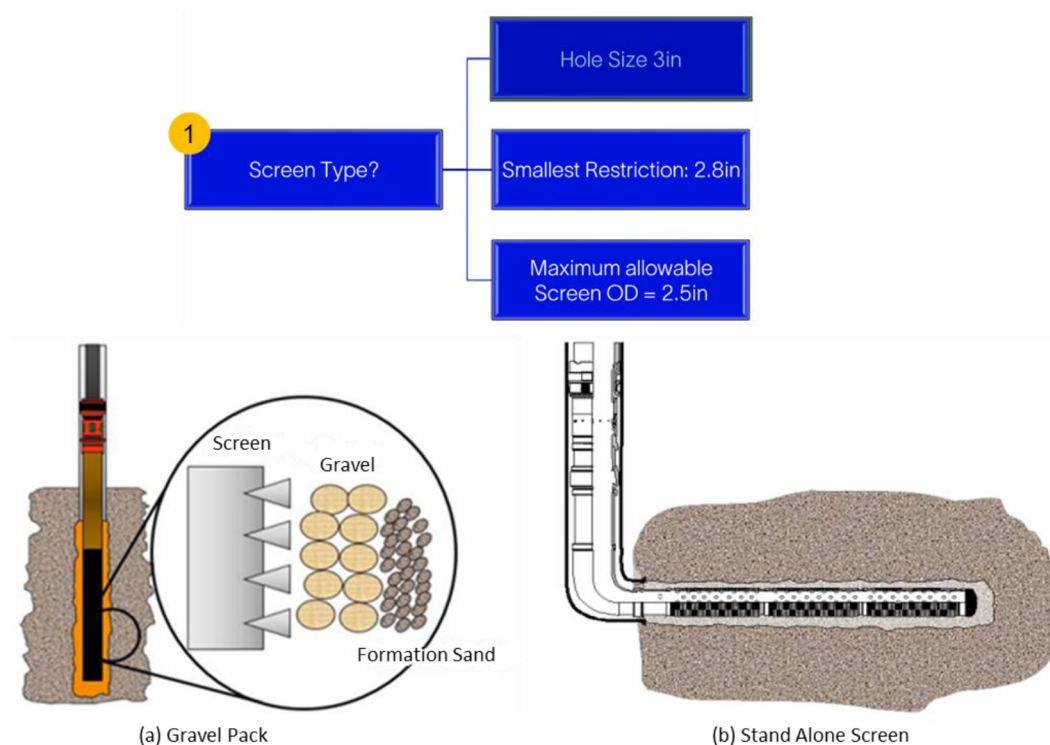
- The existing upper completions consist of 3-<sup>1</sup>/<sub>2</sub>" production tubing, which significantly limits the size and configuration of lower completions.



- To pass through the smallest restrictions (~2.813" ID), the lower completion design was limited to slim configurations (typically 2- $\frac{1}{2}$ " OD).
- High dogleg severity (DLS) from CTD sidetracks, required lower completion accessories and thread connections capable of withstanding high stresses (e.g. tensile, compressive, shear) during deployment and production.

### ***Sand Control Method Selection.***

- The field has a history of sand production that caused premature equipment failures and productivity decline in earlier wells, making effective sand control mandatory to ensure sustainable production and long-term asset integrity.
- Openhole Stand-Alone Screen (OHSAS) was chosen as the preferred method due to its reliable sand exclusion and simplified deployment, as conventional gravel pack methods were considered unsuitable due to high friction losses and limited pump capacity in horizontal slimhole environments.
- Screen sizing was based on extensive Particle Size Distribution (PSD) analysis and Sand Retention Test (SRT), where 19" premium sintered mesh screen was selected for its high mechanical strength and efficient sand retention in slimhole completions.



**Figure 6—Sand Control Option**

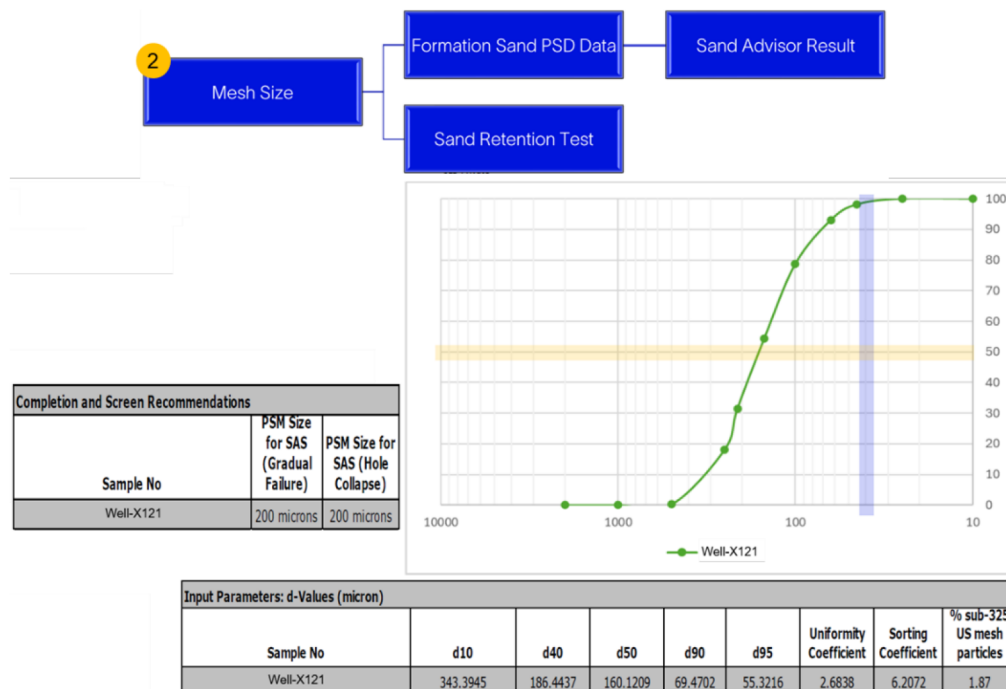


Figure 7—Screen Sizing

### ***Completion Assemblies Configuration.***

- Field X is complex multi-layered sandstone reservoir interbedded with shale and coal layers, extensive faulted and compartmentalized.
- Swellable packers were incorporated into the lower completion to isolate non-productive zones and prevent crossflow, selected for its reliability and ease of installation in horizontal slimhole completions.
- All completion assemblies were validated for compatibility with drilling/completion fluids, mechanical stresses and longterm production conditions.

### ***Erosional Velocity Validation.***

- Erosional Velocity Ratio (EVR) simulations were performed using advanced well modelling and API 14E methodology ( $C = 100$  for L80 tubulars), to identify the risk of erosion and to select the suitable sand control method.
- The simulations confirm that  $EVR < 1$  for all wells, with Fluid Mean Velocities (FMV) ranging from 1 to 15 ft/s, indicating minimal erosion risk throughout the completion life.

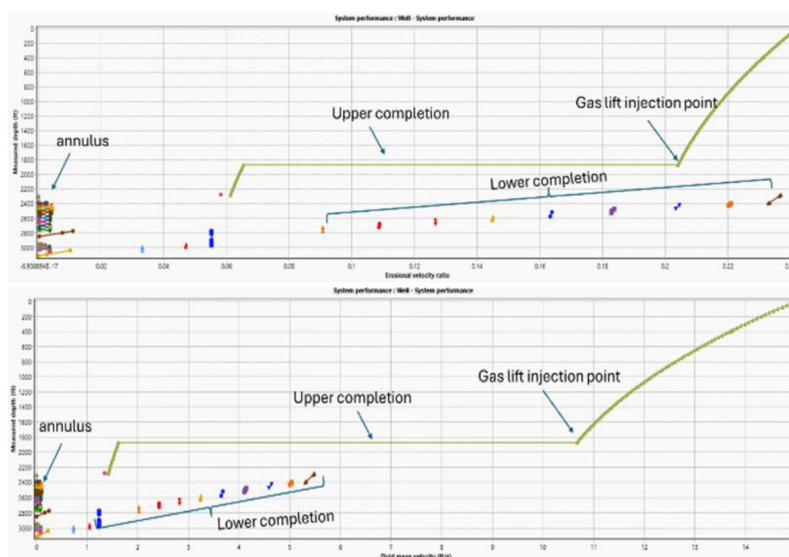


Figure 8—Erosional Velocity Ratio and Fluid Mean Velocity Simulations

## Method of Operations

The methodology for executing CTD in Field X was heavily influenced by regional experience and lessons learned from earlier campaigns, particularly the 2011 pilot phase and the 2013 SEWOP campaign. Two main operational modes have historically been applied in Malaysia and the wider South China Sea:

1. **Platform-Based CTD Operations** - feasible only when the host platform has adequate deck space, structural capacity and crane lifting capability to accommodate the full CTD spread and personnel. This method is cost-effective but limited to larger, newer platforms.
2. **Liftboat/SEWOP Assisted Operations** - as demonstrated in the 2013 campaign (Ooi et al., OTC-26690), a SEWOP can provide additional deck space, accommodation and stable lifting capacity when host platforms are constrained. This approach minimized modification to aging platforms and reduced weather dependency but incurred higher logistics and mobilization costs (i.e. similar as bringing jack-up rig to the platform).

In Field X, platforms were small with restricted crane capacities and limited space. Replicating the SEWOP approach was technically feasible but not cost-efficient for the scale of the campaign. Instead, the project team adopted a CTD catenary system supported from an Accommodation Work Barge (AWB), an evolution of prior practices that maintained the logistical advantages of SEWOP while reducing overall cost and complexity.

The catenary method involved deploying the CTD reel and pumping equipment on the AWB, with the CT suspended in a controlled catenary configuration from the vessel to the platform wellhead. This eliminated the need to lift heavy CTD reels onto the platform, reduced crane load requirements and provided a safer operation in adverse offshore conditions. Importantly, it also shortened rig-up and rig-down times compared to liftboat-based operations.

The sequence of CTD offshore operations is a crucial factor for cost-effective campaign. Building on the previous CTD campaigns, it has become evident that separating the well preparation and the actual CTD operation is the most effective strategy for reducing total costs. This approach is necessary due to the complexities involved in well preparation, especially since older, idle wells may present unpredictable challenges.

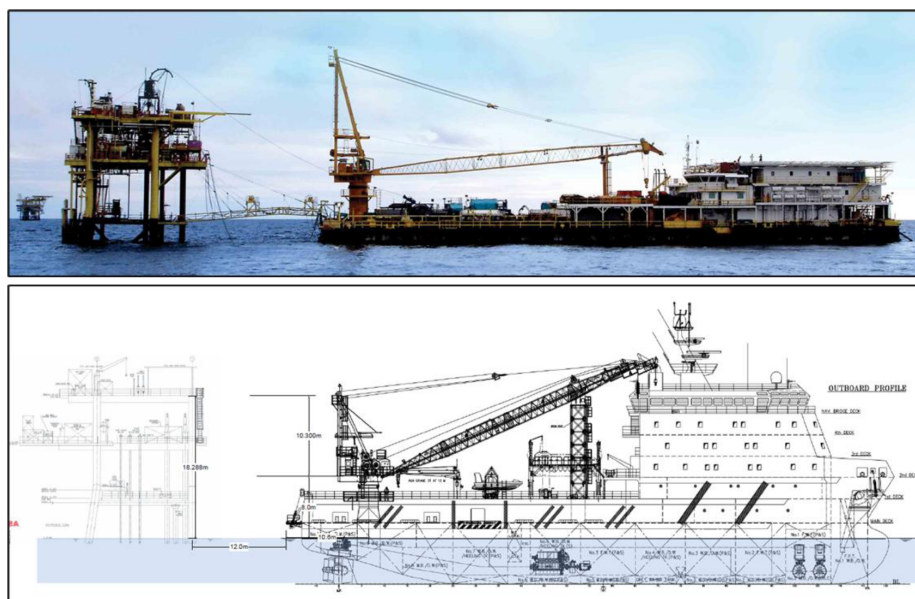


Figure 9—CT Catenary Operation

### *Pre-CTD Well Preparation.*

- Site visit to perform risk assessment and condition of the platform.
- Well integrity validation (e.g. pressure tests, drift run, caliper log, CCL).
- Isolating the existing perforations with cement or mechanical barrier.
- For exiting above the packer, a cement packer job is required to fill the space between the production tubing and casing, to allow the window milling and drilling assembly to safely pass through the tubing and casing while maintaining integrity during drilling and later in production.
- Whipstock setting for sidetrack kick-off.

### *Pre-CTD Well Preparation.*

- Dual exit window milling (i.e. tubing and casing).
- Drilling of build section (i.e. up to 45deg/100ft).
- Drilling of lateral section (i.e. up to 3,000ft openhole) with geosteering support.
- Openhole formation evaluation via CT e-line logging.
- Deployment of lower completion (e.g. RIH liner, OHSAS, swellable packers, etc.).
- Well kick-off using N2 lifting or fluid displacement.

## Conclusions and Future Outlook

CTD has proven to be an effective and fit-for-purpose technology for unlocking bypassed reserves in marginal fields particularly where production decline and a high proportion of idle wells challenge the economics of conventional drilling. Despite the inherent complexities of offshore CTD operations, ranging from logistical constraints to execution risks, it remains a viable and often advantageous alternative drilling methods when carefully planned and executed.

Although CTD has been successfully implemented in previous campaigns offshore Malaysia, its application in Field X marks a significant milestone being the first deployment after a decade-long hiatus. This project highlights the importance of meticulous planning and robust workflow to ensure successful CTD operation. Key elements include rigorous candidate selection, optimized trajectory design, tailored fluid systems and reliable completion strategies, plays a critical role in determining the overall technical and commercial outcome of CTD campaign.

Equally important is the integration of multidisciplinary expertise and seamless collaboration between the operator, service providers, subject matter experts (SME) and operations teams. A cohesive cross-functional approach fosters better communication, minimizes operational risks, enhances decision-making and ensures alignment with project objectives from planning to execution.

The CTD initiative in Field X not only demonstrates the capability to safely and efficiently revitalize production from mature offshore assets but also establishes a benchmark for future applications in Malaysia and the wider region. As Malaysia accelerates toward its Net Zero Emissions target by 2050, it becomes increasingly important to deploy efficient, lower-carbon oilfield technologies where continued hydrocarbon production remains necessary. By combining technical precision with collaborative execution, CTD offers a sustainable pathway to extend field life, maximize recovery and support long-term energy resilience. This will ensure CTD continues to evolve as a proven enabler for marginal field redevelopment and production rejuvenation cross the region.

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